Self Distillation using only supervised fine tuning: a duplex cascade case study

Felipe Penhorate Carvalho da Fonseca

***Abstract*—Fine-tuning a full-duplex speech model to emit conversational turn-taking tags (the *duplex cascade* micro-turn protocol) with plain supervised fine-tuning (SFT) regresses the model's general capabilities: function calling, instruction following and world knowledge degrade because the off-policy training signal pushes the whole next-token distribution toward tag-heavy dialogue. We show that this regression can be avoided *without reinforcement learning and without on-policy rollouts* by borrowing the self-distillation idea of *Self-Distillation Fine-Tuning* (SDFT): the frozen original model acts as its own teacher, and the fine-tune optimizes a two-term loss—the weighted cross-entropy on the duplex tags exactly as before, plus a per-token forward-KL to the frozen base model on every other (content) position. The content term *anchors* the student to the pretrained distribution everywhere the tags are not explicitly supervised, which is precisely the mechanism that prevents regression. We train a QLoRA adapter on 10k teacher-derived duplex dialogues and evaluate against both the base model and the plain-SFT baseline. The distill model learns the protocol (96.4% tag accuracy) while keeping function calling and general instruction-following at base level (100% vs the SFT baseline's 55.6% and 41.7%), and it stays within 1–2 pp of base on MMLU, HellaSwag, ARC and TruthfulQA, and above base on IFEval—where the SFT baseline drops broadly. The only measured regression is on multi-step math reasoning (GSM8K). These results establish per-token KL-anchoring to the frozen base as a practical, RL-free anti-forgetting mechanism for task-specific fine-tuning.**

Because teacher-forced tag accuracy is blind to whether the protocol is actually *used* when generating, we add a free-generation evaluation built on Full-Duplex-Bench and run it on all three variants trained in this project plus the upstream DuplexCascade release, which we re-measure with the same harness and which reproduces its own published turn-taking numbers (ATA 0.896 vs. the published 0.858). All three variants beat that published accuracy, but the comparison exposes what teacher forcing hid: the distill model is competitive on turn-taking accuracy yet has by far the weakest interruption relevance (1.59 vs. 3.56 and 4.36) and commits to the floor $13\times$ more slowly (3.29 s vs. 0.25 s), because the content-KL anchors the teacher's generation style along with its capabilities, and the teacher occasionally autocompleted the user's turn. The same harness also identifies a property of the shared training data rather than of any model—the system-backchannel tag occurs zero times, so that dimension is not testable—and shows that the assistant micro-turn length distribution, which differs between the two generations of the source dialogues, moves the duration-gated turn-taking metric itself.

***Index Terms*—self-distillation, continual learning, supervised fine-tuning, full-duplex dialogue, turn-taking, evaluation, full-duplex benchmark, catastrophic forgetting, function calling**

I. Introduction

Large language models that power interactive systems are often post-trained on narrowly defined behavioral protocols. A representative example is the *duplex cascade* paradigm [2]: a conversational LLM is fine-tuned to emit special turn-taking tokens ("<|user is speaking|>", "<|user finish speaking|>", "<|user interruption|>", etc.) that drive a VAD-free, full-duplex speech pipeline. Learning the protocol is easy—the supervision is cheap, synthetic and deterministic—but the fine-tune is destructive. In our pt-BR implementation of the duplex cascade, the plain-SFT model that learns the tags with 98% accuracy simultaneously loses most of its ability to call functions (100% $\rightarrow$ 55.6%) and follow general instructions (100% $\rightarrow$ 41.7%). This is catastrophic forgetting in its most practical form: the model becomes a good turn-taker and a bad assistant.

The conventional remedy is to make the post-training signal *on-policy*, via reinforcement learning (RL) with an explicit reward [7], [8]. But rewards are often unavailable, and on-policy RL (PPO/GRPO) is expensive and brittle. A recent line of work, *Self-Distillation Fine-Tuning* (SDFT) [1], obtains the benefits of on-policy learning from demonstrations alone: the model is used in two roles—a *teacher* conditioned on a demonstration, and a *student* conditioned only on the query—and the training minimizes a KL divergence between the two on the student's own trajectories. SDFT substantially reduces catastrophic forgetting compared to SFT, but it still requires on-policy rollouts (roughly $4\times$ the wall-clock of SFT).

We show that for the duplex-cascade setting there is a much simpler instantiation that needs *neither RL nor rollouts*. Our key observation is that the "new skill" (the turn-taking protocol) and the "everything else" (the general capabilities we want to preserve) are *separable by token position*: the tags live at supervised positions, the content lives at the other positions. We therefore split the objective in two:

- **Tags**: the weighted cross-entropy to the supervised tag tokens, exactly as in the plain SFT recipe.
- **Everything else**: a per-token forward-KL to the *frozen base model* on content positions, so the student's distribution is anchored to the pretrained policy wherever there is no explicit tag supervision.

The teacher is the frozen original model itself (conditioned on the same duplex sequence via the tag tokens in its input), so no second model and no rollouts are needed. Because the student is *base + LoRA*, the teacher forward is obtained with zero extra VRAM by temporarily disabling the adapters. We precompute the teacher's top-k next-token logits once (offline cache), which removes the second forward pass entirely and

makes the run no more expensive than SFT.

We evaluate on a pt-BR duplex-cascade model (Qwen3-4B-Instruct-2507, QLoRA, 10k teacher-derived dialogues). The resulting *distill* model matches the plain-SFT model on the protocol (96.4% vs 98% tag accuracy) while retaining base-level function calling, general instruction following, MMLU, HellaSwag, ARC and TruthfulQA, and *improving* on IFEval. The plain-SFT baseline regresses on all of them. The one caveat is GSM8K, where the distill model drops 11 pp below base—a limitation we discuss in Section VI.

Is the protocol actually used? Every number above is teacher-forced: at each supervised position the model is handed the correct prefix and only its next-token choice is scored. That measures whether the protocol was *learned*; it says nothing about whether the model *decides* correctly when it generates on its own, which is what a deployed duplex system needs. We therefore build a free-generation evaluation on Full-Duplex-Bench [15] and run it on the three variants trained in this project as well as on the upstream DuplexCascade release, re-measured with the same harness rather than quoted. Three findings follow, and they refine the picture above. (i) The harness is trustworthy: the upstream release lands on its own published turn-taking numbers. (ii) All three variants exceed the published turn-taking accuracy—but the distill model is the worst of the three once reply *quality* and *latency* are counted, which is exactly the cost teacher-forced tag accuracy cannot see. (iii) A property of the data, not of any model, makes one FDB dimension untestable here: the system-backchannel tag never occurs in training. We also document the measurement trap that dominated our own first attempt—the evaluated input stream must match the training stream, because feeding whole sentences instead of ASR-sized pieces changes the reported turn-taking accuracy by a factor of four.

II. Background and Related Work

A. Catastrophic forgetting and off-policy fine-tuning

Sequential SFT is inherently off-policy: the model learns to imitate expert actions on a fixed offline distribution, and prior work has shown this leads to severe forgetting when the model is adapted to a new task or domain [4]–[6]. On-policy methods—training on trajectories induced by the model itself—forget substantially less [3], [7]–[9]. DAgger [3] established that on-policy imitation avoids the compounding-error failure mode of behavior cloning. More recently, "SFT memorizes, RL generalizes" [7] and related analyses [8] showed empirically that RL-based post-training retains general capabilities far better than SFT.

B. Self-distillation

Our method is most closely related to *Self-Distillation Fine-Tuning* (SDFT) [1]. SDFT builds the teacher by conditioning the model on an expert demonstration, $\pi(\cdot|x, c)$, and trains the student $\pi_\theta(\cdot|x)$ to match it on the student's own rollouts:

$$\begin{aligned} L(\theta) &= D_{KL}\big(\pi_\theta(\cdot|x) \,\|\, \pi(\cdot|x, c)\big) \\ &= \mathbb{E}_{y\sim\pi_\theta} \sum_t \log \frac{\pi_\theta(y_t|y_{<t}, x)}{\pi(y_t|y_{<t}, x, c)} \,. \end{aligned} \tag{1}$$

The paper shows that on Tool Use, Science Q&A and Medical skill-learning, SDFT achieves higher new-task accuracy than SFT while keeping previous-capabilities performance near the base model, and that an EMA of the student is the most stable teacher. Our work differs in two ways. First, we do *not* run on-policy rollouts; instead the tags are supplied by a deterministic construction rule, so the only on-policy-sensitive component is the content, which we anchor to the frozen base distribution directly. Second, our teacher is the frozen base model (not a demonstration-conditioned variant), which is exactly the "keep everything else the same" reference the anti-forgetting objective requires.

Context distillation [11], [12] also uses a model conditioned on extra context as a teacher for the model without it, but typically distills offline from static contexts and does not combine a separate supervised tag term with a content-anchoring KL. *Learning by distilling context* [12] distills a prompt-conditioned teacher into the base model; our tag term is the analogous "hard" supervision and our content term the distillation.

C. Parameter-efficient fine-tuning

We train with QLoRA [14] (4-bit NF4 base, LoRA [13] rank 16, 16 α, dropout 0) via the Unsloth [16] training library, keeping the embedding and LM head trainable so the new special-token rows can be learned.

III. Method: DuplexCascade-Distill

A. Setup

Let the base model be π_0 (frozen) and the student π_θ be the same weights plus a LoRA adapter (θ = adapter parameters). The training corpus is a set of *duplex sequences* x produced by the duplex micro-turn builder: a dialogue is split into micro-turns and conversational special tokens ($\mathcal{T}$) are inserted deterministically at turn boundaries (e.g. "<|user finish speaking|>" before every system turn, probabilistic interruptions/backchannels). Every position t of x is thus either a *tag* position (label $y_t \in \mathcal{T}$) or a *content* position (label y_t is ordinary text). User-side and header positions are masked (label $-\infty$), as in the original recipe.

B. Teacher content

The tags themselves come from the deterministic builder. The *content* comes from the teacher: the frozen base model, prompted with a *duplex-cascade trigger* system message ("you are a DuplexCascade full-duplex voice assistant… speak in short, natural micro-turns… no tags"), generates each assistant turn conditioned on the history. At first the base model does not follow the protocol faithfully—that is expected and irrelevant; we keep only its *content* and let the deterministic

builder place the tags. This yields the aligned demonstration sequence x.

C. Loss

Let T be the set of tag positions and C the set of content positions. The student loss is

$$L(\theta) = \underbrace{\sum_{t \in T} w_t \, \mathrm{CE}\big(\pi_\theta(\cdot|x_{<t}),\, y_t\big)}_{\text{tags: as in plain SFT}} + \lambda \underbrace{\sum_{t \in C} D_{KL}\big(\pi_0(\cdot|x_{<t}) \,\|\, \pi_\theta(\cdot|x_{<t})\big)}_{\text{content: anchor to the frozen base}}, \tag{2}$$

where w_t are the per-special-token weights of the original recipe (e.g. "<|user_finish_speaking|>"= 10, "<|user_interruption|>"= 5. The KL is in the forward direction and computed in cross-entropy form $-\sum_k p_0(k) \log p_\theta(k)$ over the teacher's top-k (k=32) next-token support, with the student logits gathered at those ids. Because the teacher is conditioned on the same x (its input already contains the tags), its content distribution is the natural "what would the original model say in a duplex dialogue" reference. The single hyper-parameter λ trades tag fidelity against anchoring; we use λ=1.0.

D. Practical implementation

- **Teacher forward is free.** The student is base + LoRA, so the teacher is the same weights without adapters: one additional `no_grad` forward under `disable_adapter()` costs zero extra VRAM.
- **Offline teacher cache.** To avoid the second forward in the training loop (and the GPU-allocator pressure it creates at the 16 GB ceiling), we precompute the teacher's top-k ids/logits for every position of every sequence once (one bf16 forward pass over the corpus, ≈36 min). Training then reads cached targets and runs at the plain-SFT cost. This is the difference between our ∼10 s/step and the two-forward path.
- **Anchor set (D2 variant).** A small set of general-capability prompts (function calling, QA) is distilled the same way, pinning the LoRA delta to ≈ 0 outside the duplex protocol.
- **Artifact masking.** Following the SDFT "learned artifacts" fix [1], the first few content tokens after each "<—user finish speaking—¿" tag are left unsupervised so the student does not copy teacher surface artifacts.

IV. Experimental Setup

A. Models

Base: `Qwen/Qwen3-4B-Instruct-2507` (dense Qwen3, 151.9k vocab). Student: QLoRA (r= 16, α=16, dropout 0, all linear modules, trainable embeddings and LM head), 7 duplex special tokens appended and Gaussian-initialized. Teacher: the same base, frozen. All models evaluated as bf16.

B. Data

10,000 pt-BR dialogues from the existing corpus had every assistant turn regenerated by the teacher (trigger prompt, temperature 0.7, top-p 0.95), then passed through the deterministic micro-turn builder (user chunks 1–7 tokens, system chunks 10–48, probabilistic pauses/interruptions/backchannels), yielding 10,249 training sequences at $\leq$1024 tokens. 300 anchor prompts (150 function calling, 150 general QA).

C. Training

2,000 optimizer steps, batch 1 × grad-accum 16 (effective 16), AdamW lr 10^{-4} (embeddings 2×10^{-4}), 20-step warmup, bf16, gradient checkpointing, λ=1.0 (chosen by a short sweep on $\lambda \in \{0.1, 0.3, 1.0\}$ using tag accuracy and KL drift). The `sft` baseline is the existing DuplexCascade-PT QLoRA finetune (same data family, no KL term).

D. Evaluation

Protocol: tag accuracy (argmax = supervised tag) on a held-out duplex eval split. *Capabilities*: 18 pt-BR function-calling prompts (native Qwen tool calls, greedy) and 12 general instruction/QA prompts. *Distribution drift*: mean per-token forward-KL (student ‖ frozen base) on the content positions of the held-out split—a direct measure of "everything else stayed the same". *Standard benchmarks*: HellaSwag, ARC-Challenge, MMLU (2000-example subsets), GSM8K (5-shot, 300), TruthfulQA (mc2), IFEval (40) via lm-evaluation-harness, raw prompts, no chat template, identical configs for all three models.

E. Turn-taking in free generation (Full-Duplex-Bench)

The tag-accuracy metric of Section IV is teacher-forced: at every supervised position the model is handed the correct prefix and only its next-token choice is scored. That is the right measure of whether the protocol was *learned*, but it says nothing about whether the model *uses* it when generating on its own, which is what a deployed duplex system needs. We therefore add a free-generation evaluation modelled on Full-Duplex-Bench (FDB) [15].

Dimensions and metrics. FDB v1.0 scores four behaviours: *pause handling* and *backchannel* (the model must stay silent or briefly acknowledge, TOR ↓; backchannel frequency ↑ and a JSD against a ground-truth backchannel-time distribution ↓), *smooth turn-taking* (take the floor as soon as the user's turn ends, TOR ↑, latency ↓) and *user interruption* (yield, then answer the interrupting turn: TOR ↑, latency ↓, and an LLM-judged 0–5 relevance of the post-interruption reply). We use FDB's thresholds verbatim (a model utterance is a formal turn at $\geq$1 s or >3 words; a backchannel at $\leq$3 s and $\leq$3 words) and its interruption rubric verbatim.

Text-domain adaptation. Our cascade decides turn-taking in text, so the harness replays the ASR stream as timed events rather than audio and converts text to seconds with a calibrated speaking rate (3.25 words/s, measured against the pocket-tts voice the live bridge uses and anchored to the rate implied by FDB's own released transcripts). Latency

is measured as the model's real inference compute between the user's turn ending and the model committing, which is the honest text-domain analogue of acoustic response latency and is what separates a tag-emitting LLM from a separate policy companion. Interruption relevance is judged by a local `Qwen3.5-9B` GGUF under FDB's own rubric; the judge is not GPT-4o, so that column ranks our models against each other rather than against the published absolute. All runs are greedy and every model is served through the same llama.cpp stack at the same quantisation.

The input stream must match training, or nothing else matters. The models never see whole sentences. The micro-turn builder delivers each user turn as 1–7-**token** pieces (Section IV), so we re-use that algorithm verbatim—including its mid-word cuts and leading-space artefacts—to generate the evaluated stream. This is not a detail. Our first harness fed whole sentences, on which a complete sentence is an unambiguous "the user has stopped" cue; the turn-end decision became artificially easy, pause-handling TOR looked excellent (0.015) and the numbers were meaningless. With the stream restored, the same model scores 0.080. Measured on 3,000 training records the stream averages **2.9 words per user piece** (median 3, p90 5) arriving as 3.7 pieces per turn.

Turn size on both sides, and why it is not a constant. The source dialogues exist in two generations. The original set instructed *"respostas curtas do Assistente. Nada de textos longos"*, and the chunker fixed every assistant micro-turn at exactly 10 tokens; the resulting models spoke only in $\sim$10-token bursts. The continuation set (the data both models here are trained on) mixes *short* (weight 6), *long* (2) and *mixed* (2) styles, so a fraction of assistant turns are 2–4-sentence replies up to 220 words, and system chunking samples 10–48 tokens instead of a constant. The consequence for evaluation is a mixture, not a mean: measured over the training data, an assistant content burst averages 6.1 words but has median 5, p90 12 and max 37, with 50% of bursts at ≤ 5 words and 1% above 20. Because FDB's TOR gates are *durations* ($\geq$1 s, $\leq$3 s), that heavy right tail is not cosmetic—a model trained on longer bursts crosses the "formal turn" threshold with fewer words, so the assistant-side length distribution moves the very metric being reported. The user side is correspondingly fine-grained (2.9-word pieces), and a scenario generator that joins several sentences per turn—as ours first did, producing 35–71-word turns with 2–4$\times$ the sentence-final ambiguity of the real data—reverses the ranking of the models. We therefore generate every task's turns to match the measured distribution, and we report the per-side lengths with the results.

V. Results

A. Protocol vs. capability retention

Table I shows the central result. The *distill* model learns the protocol (96.4% tag accuracy, only 1.6 pp behind the SFT baseline) while keeping function calling and general instruction-following at *base level* (100%), and a KL drift only slightly above the base floor. The plain-SFT baseline, by contrast, learns the protocol (98%) but regresses catastrophically on the format-following capabilities (function calling 55.6%, general 41.7%) and drifts $\sim$3.5$\times$ further from the base distribution.

TABLE I
Protocol and capability retention (our pt-BR eval sets).

	base	**distill**	sft
Tag accuracy (%)	0.0	**96.4**	98.0
Function calling (%)	100	**100**	55.6
General instruction/QA (%)	100	**100**	41.7
KL drift vs base (nats)	0.51	1.07	4.76

TABLE II
Standard benchmarks (lm-evaluation-harness 0.4.12; subsets as noted).

Benchmark	base	**distill**	sft
HellaSwag acc-norm (n=2000)	0.584	0.566	0.593
ARC-Challenge acc-norm (n=2000)	0.584	0.563	0.574
MMLU (n=2000)	0.705	0.698	**0.660**
GSM8K exact (n=300, 5-shot)	0.890	0.777	0.827
TruthfulQA mc2	0.626	**0.649**	0.601
IFEval loose / strict (n=40)	0.730/0.651	**0.762/0.683**	0.603/0.524

B. Standard benchmarks

Table II reports the lm-evaluation-harness results. The distill model is within $\sim$1–2 pp of base on knowledge (MMLU), commonsense (HellaSwag) and reasoning (ARC), *above* base on factuality (TruthfulQA $+2.3$ pp) and instruction following (IFEval loose $+3.2$ pp), and at base on the format-following evals of Table I. The SFT baseline drops on MMLU (-4.5 pp), TruthfulQA (-2.5 pp), IFEval (-12.7 pp) and the custom sets. The one regression of the distill model is GSM8K (-11.3 pp), discussed next.

C. Turn-taking in free generation: results

Table III reports the FDB results for the three variants trained in this project, the untrained base as a control, and—measured with the same harness rather than quoted—the original English DuplexCascade release from `sbintuitions`, which we downloaded, merged to a 4-bit GGUF and served through the identical stack.

The harness reproduces the reference. The upstream release, measured here, lands on its own published numbers: pause handling 0.102/0.273 against the published 0.058/0.222, smooth turn-taking 0.975 against 0.832, interruption TOR 0.985 against 0.955, interruption relevance 3.92 against 4.02, and ATA 0.896 against 0.858. A re-implementation that reproduces the reference model's scores is measuring the same thing the reference measures, which is the strongest available check on everything else in the table.

All three variants beat the reference on turn-taking accuracy. They reach ATA 0.955 (distill), 0.971 (SFT) and 0.973 (MTP-companion)—every one above the upstream model's 0.896 and above the published 0.858—and all three sit at or better than the published figures on every TOR column. They

TABLE III

FULL-DUPLEX-BENCH, PT-BR (727 SCENARIOS PER MODEL; TOR AND LATENCY ↓ IS BETTER, TOR AND RATING ↑ IS BETTER; BEST TRAINED RESULT PER COLUMN IN BOLD). "ORIGINAL" IS THE UPSTREAM ENGLISH RELEASE MEASURED WITH THE SAME HARNESS; THE PUBLISHED ROW IS ITS PAPER'S ENGLISH AUDIO RESULT AND IS SHOWN AS A RANGE CHECK, NOT A BASELINE.

Model	Pause TOR_{syn}	Pause TOR_{cand}	BC TOR	Smooth TOR	Floor lat. (s)	Intr. TOR	Intr. relevance	Stop lat. (s)	**ATA**
DuplexCascade-PT (SFT)	0.080	0.032	1.000	**1.000**	**0.25**	**0.995**	3.56	**0.035**	**0.971**
DuplexCascade + MTP companion	0.073	**0.000**	1.000	0.975	0.33	0.990	**4.36**	0.066	**0.973**
DuplexCascade-Distill	0.095	0.065	1.000	0.983	3.29	0.995	1.59	0.351	0.955
Original DuplexCascade (upstream)	0.102	0.273	**0.436**	0.975	1.02	0.985	3.92	0.323	0.896
Base (no fine-tune, control)	0.000	0.000	0.000	0.000	–	0.000	–	0.15	0.000
published DuplexCascade (EN audio)	*0.058*	*0.222*	*0.218*	*0.832*	*1.72*	*0.955*	*4.02*	*1.23*	*0.858*

get there differently: the SFT model is fastest by an order of magnitude (stop latency 0.035 s, response 0.28 s) because one model does everything, whereas the companion has the best interruption relevance (4.36) but cannot cut the audio until its separate 0.8B controller has been queried (stop latency 0.066 s).

What teacher forcing hid. The distill model is competitive on turn-taking (ATA 0.955) but has by far the weakest interruption relevance (1.59 against 3.56 for SFT and 4.36 for the companion) and is $13\times$ slower to commit to the floor (3.29 s against 0.25 s). Teacher-forced tag accuracy (96.4%, Table I, 1.6 pp behind SFT) is blind to exactly this: both models put the right tag at the right position when given the correct prefix, but in free generation the distill model's continuation artifact—the known failure in which it opens by finishing the user's own sentence—degrades the reply and delays the turn. The live bridge masks it by stripping leading continuation punctuation; this evaluation does not, so the cost is visible. This is the concrete form of the exposure-bias gap between our protocol metric and deployed behaviour, and it is the strongest argument for evaluating turn-taking in free generation rather than at supervised positions.

Cross-lingual transfer. All variants are fine-tuned on pt-BR only, and the protocol transfers: ATA is 0.986 (SFT), 0.984 (companion) and 0.924 (distill) on the English scenarios, i.e. SFT and the companion are as good or better in English than in pt-BR, with interruption relevance rising to 4.88 for the companion. Only the distill variant degrades (0.955 → 0.924, interruption TOR 0.995 → 0.915), consistent with it being the most fragile of the three.

Backchannel is a data gap, not a capability gap. Every pt-BR variant scores BC TOR = 1.000 where the upstream model scores 0.436: they always take a formal turn during a monologue, while upstream backchannels. The cause is in the training data—across 10,249 sequences the tag `<|system backchannel|>` occurs **zero** times, and `<|user backchannel|>` only 1,262 times (0.09% of duplex tags)—so the token exists in the vocabulary but was never trained. That column is not evidence about these models and we do not interpret it.

One more check the table gives us. The untrained control scores `chunk_accuracy` 0.85 on the turn-end decision—*higher* than every trained model—purely by never taking the floor. Our averaged turn-taking accuracy gates on the model demonstrably taking turns elsewhere and scores an unmeasured interruption as zero, which puts the control at 0.000.

VI. DISCUSSION AND LIMITATIONS

Why the content-KL works. The tags live at supervised positions and are predicted with the original high weights, so they are learned essentially as before. Every other position is anchored to the frozen base, which is exactly the behavior we want to preserve. The mechanism is a single-stage, per-token analogue of the "re-invoke" two-stage procedure [10] and of SDFT's on-policy anchoring [1], but it requires neither rollouts nor a reward.

The GSM8K caveat. The distill model drops 11 pp below base on GSM8K, a long-form multi-step reasoning task. This is the honest tradeoff of a strong anchoring term (λ=1.0): content distributions are heavily pinned to the base, and on 300 samples the 4B base's fragile chain-of-thought is sensitive to any distributional shift. Two natural follow-ups are (i) a lower λ (our sweep did not include a math benchmark), and (ii) the EMA-teacher or on-policy variants of SDFT (SPEC D3/D4), which track the student's progress and may preserve the student's adaptability on reasoning while still reducing forgetting.

Free-generation cost of the anchoring term. The content-KL preserves capabilities (Tables I and II) but, at λ=1.0, it also anchors the model to the teacher's content *generation style*, and the teacher was occasionally autocompleting the user turn when its auxiliary trigger asked for a short answer. The distillation therefore inherits that continuation artifact, which shows up in Table III as the worst interruption relevance (1.59) and the slowest floor commitment (3.29 s) of any model measured, even though its teacher-forced tag accuracy is 96.4%. The fix we planned is at the data level, not the loss: our SPEC's §6 R0 regenerates the teacher content with the corrected *v2* trigger ("responda de forma natural e completa … não se repita") and retrains, so the artifact is absent from the anchored distribution rather than stripped at inference by the bridge. The FDB harness now provides the measurement that tells us whether that retrain succeeded: relevance and stop latency, not tag accuracy.

A. Next steps: learning only part of the answer

The crux of our method is the *positional separation* of the objective: a small, well-localized set of supervised po-

sitions carries the new behavior, and every other position is anchored to the frozen base. This is a special case of a much more common and important fine-tuning pattern—*partial-answer supervision*—where the student's target output is already largely correct and only a small, identifiable part must be edited. Our approach should transfer directly to any such setting, and we plan to evaluate it there next. Concretely:

- **Guardrail classifiers.** The most immediate target is fine-tuning an LLM to behave as a *classifier*. For a content-moderation guardrail, the input is a text and the target output is a verdict from a small fixed taxonomy (e.g. "OK" / "FLAGGED", or a category label), often preceded by a one-line justification. Plain SFT on this data collapses the model's output distribution onto the label tokens and erodes general abilities. With our recipe, the *verdict position is the "tag"* (weighted cross-entropy) and the justification and everything else are the "content" (forward-KL to the frozen base). The model then learns the decision boundary while keeping its language, instruction-following and knowledge intact—exactly the retention we measured for the duplex tags (Table I).
- **Other candidate settings.** (i) *Schema/format fix-ups* where the answer semantics are correct but the formatting (JSON keys, tool-call syntax) must change; (ii) *claim correction* (medical, legal, RAG) where only a factually wrong token span needs editing; (iii) *code repair* where only the buggy lines change; (iv) *citation insertion*.

The enabling requirement is that the "new behavior" is locatable at predictable token positions. When it is—a verdict token, a tag, a format delimiter—the off-policy recipe above is directly applicable and stays at plain-SFT cost. When the edit location is *not* known a priori, one can first align the student to a canonical template (exactly as the deterministic micro-turn builder does for duplex dialogue) so the edit collapses onto predictable positions; otherwise the on-policy SDFT variant [1] is the fallback. This planned evaluation also speaks to the principle of *RL's Razor* [9]: forgetting is governed by the KL to the base on the new task, and partial-answer settings are precisely where an explicit per-token KL constraint is cheapest and most natural, because the supervised support is small.

Limitations. This is work in progress: our evaluation is on a single backbone (Qwen3-4B), a single language (pt-BR) and subset-scale benchmarks; the IFEval and GSM8K numbers carry per-sample stderr of order 2 pp. We did not ablate λ against math, nor test the full 50k-scale data regime; we plan to evaluate with substantially more data in future work. The offline cache assumes the teacher is fixed, which is true by construction but means the student can never "teach forward" a better protocol—acceptable here because the tags are deterministic.

VII. Conclusion

We showed that the duplex-cascade protocol can be learned with plain supervised fine-tuning *without* the catastrophic regression that usually accompanies it, by splitting the objective into a supervised tag term and a per-token forward-KL to the frozen base model. The approach needs neither reinforcement learning nor on-policy rollouts, and with an offline teacher cache it costs no more than SFT. On a pt-BR duplex-cascade model it matches the plain-SFT protocol accuracy while preserving base-level function calling, general instruction following, MMLU, HellaSwag, ARC and TruthfulQA and improving IFEval—where the SFT baseline regresses broadly. Per-token KL-anchoring to the frozen base is a practical, RL-free anti-forgetting tool that generalizes beyond duplex dialogue.

Free-generation evaluation qualifies that conclusion in a way we think generalizes with it. Anchoring the content distribution preserves capabilities, but it also preserves the teacher's *generation style*, including the one artifact we know it has: the teacher sometimes autocompleted the user's turn, and the student inherited it. Teacher-forced accuracy could not detect this—it stayed at 96.4%—while a turn-taking benchmark measures it directly as the worst interruption relevance and a $13\times$ slower floor commitment of any model tested. The practical lesson is that a protocol metric should be paired with a free-generation metric whenever the protocol is *used* rather than merely reproduced, and that the fix for a distilled artifact belongs in the teacher data (our corrected trigger and retrain) rather than in inference-time cleanup. Two further observations are about benchmarks rather than models, and both cost us real time before we found them. A generative benchmark inherits the training stream: evaluate the same model on ASR-sized pieces and on whole sentences and its turn-taking accuracy moves from 0.08 to 0.29, so the input distribution is an experimental variable that must be reported. And a metric can be unanswerable by construction: no model here can score the backchannel dimension, because the tag it requires never appears in the training data.

Use of Generative AI

This manuscript was prepared with the assistance of generative artificial intelligence. All experimental results and the associated code were reviewed and verified by the author.

References

[1] I. Shenfeld, M. Damani, J. Hübötter, and P. Agrawal, "On-policy self-distillation fine-tuning (SDFT): self-distillation enables continual learning," arXiv:2601.19897, 2026.

[2] J. Yang, Y. Fujita, and Y. Sudo, "DuplexCascade: Full-duplex speech-to-speech dialogue with VAD-free cascaded ASR–LLM–TTS pipeline and micro-turn optimization," arXiv:2603.09180, 2026.

[3] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. AISTATS*, 2011.

[4] J. Kirkpatrick *et al.*, "Overcoming catastrophic forgetting in neural networks," *PNAS*, vol. 114, no. 13, pp. 3521–3526, 2017.

[5] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE TPAMI*, vol. 40, no. 12, pp. 2935–2947, 2018.

[6] L. Mecklenburg *et al.*, "Injecting new knowledge into large language models via supervised fine-tuning," arXiv:2404.00213, 2024.

[7] T. Chu, Y. Zhai, J. Yang, S. Tong, S. Xie, D. Schuurmans, Q. V. Le, S. Levine, and Y. Ma, "SFT memorizes, RL generalizes: a comparative study of foundation model post-training," in *Proc. ICML*, 2025, pp. 10818–10838.

[8] R. Agarwal, N. Vieillard, Y. Zhou, P. Stańczyk, S. Ramos, M. Geist, and O. Bachem, "On-policy distillation of language models: learning from self-generated mistakes," in *Proc. ICLR*, 2024, arXiv:2306.13649.

[9] I. Shenfeld, J. Pari, and P. Agrawal, "RL's razor: why online reinforcement learning forgets less," in *Proc. ICLR*, 2026, arXiv:2509.04259.
[10] K. Lu and Thinking Machines Lab, "On-policy distillation," Thinking Machines Lab: Connectionism, DOI 10.64434/tml.20251026, 2025.
[11] Y. Bai *et al.*, "Constitutional AI: harmlessness from AI feedback," arXiv:2212.08073, 2022.
[12] C. Snell, D. Klein, and R. Zhong, "Learning by distilling context," arXiv:2209.15189, 2022.
[13] E. J. Hu *et al.*, "LoRA: low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
[14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: efficient finetuning of quantized LLMs," in *Proc. NeurIPS*, 2023.
[15] G.-T. Lin, J. Lian, T. Li, Q. Wang, G. Anumanchipalli, A. H. Liu, and H.-y. Lee, "Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities," *arXiv preprint arXiv:2503.04721*, 2025.
[16] Unsloth AI, "Unsloth: 2x faster free finetuning," https://unsloth.ai, 2026.